(3) There is only one absolute maximum (and one absolute minimum) but there can be more than one local maximum or minimum.

## Points of Inflection

**Definition 7.9**

The points where the graph of the function changes from **"concave up to concave down"** or **"concave down to concave up"** are called the **points of inflection of** $f(x)$.

**Theorem 7.12 (Test for Points of Inflection)**

(i) If $f''(c)$ exists and $f''(c)$ changes sign when passing through $x=c$, then the point $(c, f(c))$ is a point of inflection of the graph of $f$.

(ii) If $f''(c)$ exists at the point of inflection, then $f''(c)=0$.

### Remark

To determine the position of points of inflexion on the curve $y=f(x)$ it is necessary to find the points where $f''(x)$ changes sign. For 'smooth' curves (no sharp corners), this may happen when either

(i) $f''(x)=0$ or

(ii) $f''(x)$ does not exist at the point.

### Remark

(1) It is also possible that $f''(c)$ may not exist, but $(c, f(c))$ could be a point of inflection. For instance, $f(x)=x^{\frac{1}{3}}$ at $c=0$.

(2) It is possible that $f''(c)=0$ at a point but $(c, f(c))$ need not be a point of inflection. For instance, $f(x)=x^4$ at $c=0$.

(3) A point of inflection need not be a stationary point. For instance, if $f(x)=\sin x$ then, $f'(x)=\cos x$ and $f''(x)=-\sin x$ and hence $(\pi, 0)$ is a point of inflection but not a stationary point for $f(x)$.

### Example 7.57

Determine the intervals of concavity of the curve $f(x)=(x-1)^3\cdot(x-5), x\in\mathbb{R}$ and, points of inflection if any.

### Solution

The given function is a polynomial of degree 4. Now,

$$\begin{aligned} f'(x) &= (x-1)^3+3(x-1)^2\cdot(x-5) \\ &= 4(x-1)^2\cdot(x-4) \\ f''(x) &= 4((x-1)^2+2(x-1)\cdot(x-4)) \\ &= 12(x-1)\cdot(x-3) \end{aligned}$$

Now,

$$f''(x) = 0 \Rightarrow x=1, x=3.$$

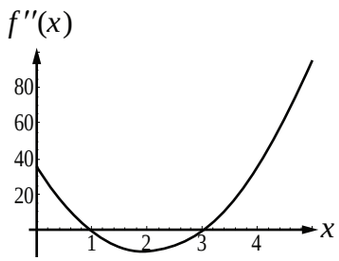



Fig.7.24



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